Exploratory Data Analysis Report

1. Global HR Key Performance Indicators (KPIs)

1,470	16.12%	237	~7 Yrs
TOTAL WORKFORCE	OVERALL ATTRITION	EMPLOYEES LEFT	AVG TENURE

Of the 1,470 employees analysed, 237 have left the company, giving an overall attrition rate of 16.12% and a retention rate of 83.88%. In simple terms, about 1 in every 6 employees has left.

2. Primary Attrition Drivers (Findings)

Bivariate analysis was conducted using Python (Pandas, Seaborn) against the target variable `Attrition`. The following factors emerged as the key drivers of employee turnover:

Factor	Finding & Statistical Observation	Impact Level
Over Time	Employees required to work continuous overtime exhibit a disproportionately high attrition volume (127 departed) compared to those who do not work overtime. Overtime acts as a severe catalyst for burnout.	**Critical**
Job Satisfaction	An inverse correlation exists between job satisfaction and attrition. Employees reporting a Job Satisfaction rating of '1' (Lowest) showed the highest attrition rate (66 departures).	**High**
Monthly Income	The average monthly income of retained employees is $6,833, whereas the average for those who left is significantly lower at $4,787. Lower-tier compensation directly correlates with higher turnover.	**High**
Commute Distance	Employees who left lived 10.63 miles from the office on average, compared with 8.92 miles for those who stayed. A longer commute adds to fatigue and resignation risk.	**Moderate**
Department & Travel	R&D (133) and Sales (92) record the most departures. Employees who travel frequently show the highest attrition rate of any travel group.	**Moderate**
Age Demographics	Younger professionals (ages 25-31) represent the largest segment of exiting employees, suggesting career mobility and entry-level churn as contributing factors.	**Moderate**

Note: These numerical findings are translated into visual representations in the Tableau interactive dashboard shown in the next section.

3. Master Dashboard Overview

The dashboard below brings all the factors together in one view, so that each employee segment can be compared against the 16.12% overall attrition rate. Blue represents employees who stayed and orange represents employees who left.

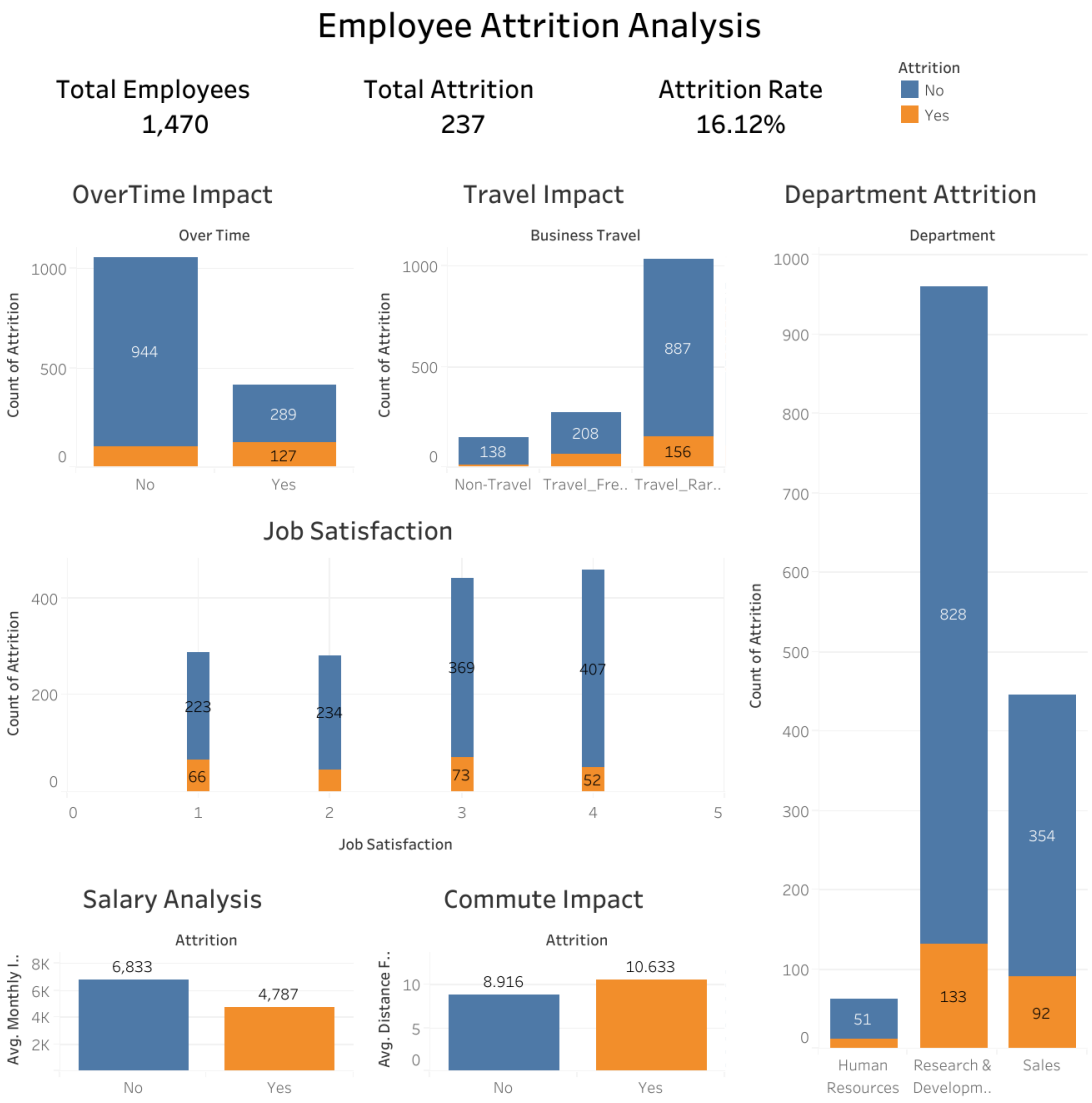


Figure 1: Employee Attrition Analysis dashboard (Tableau Public)

4. Graph-by-Graph Analysis

4.1 OverTime Impact

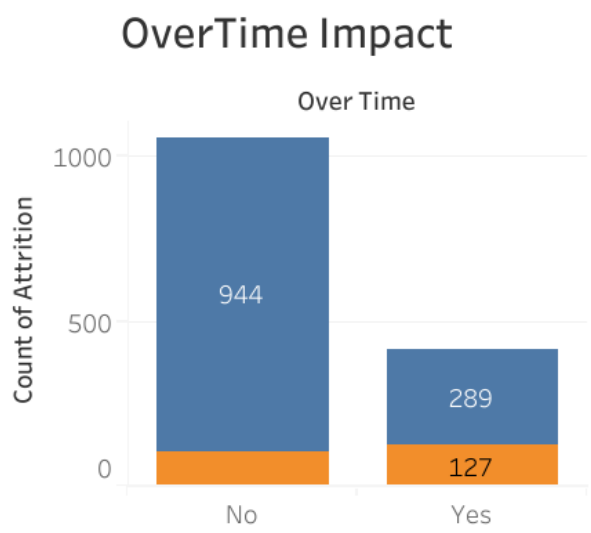


Figure 2: OverTime Impact

What it shows: The number of employees who stayed or left, split by whether they work overtime.

Finding: Overtime workers leave far more often. 127 of 416 overtime employees left (30.5%), compared with 110 of 1,054 non-overtime employees (10.4%), roughly three times the rate.

Insight: Most employees do not work overtime, but a large share of the leavers come from the overtime group. Overtime is the strongest single driver of attrition in the data.

4.2 Travel Impact

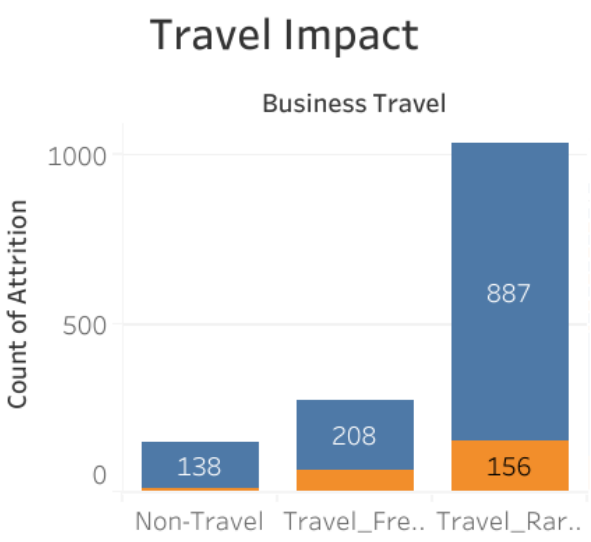


Figure 3: Travel Impact

What it shows: Attrition across Non-Travel, Travel_Frequently and Travel_Rarely employees.

Finding: Travel_Rarely employees make up most of the workforce and the highest number of resignations (156). Frequent travelers, however, have the highest attrition rate at 24.9%, compared with 15.0% for rare travelers and 8.0% for non-travelers.

Insight: The count is highest for rare travelers only because the group is large. Frequent business travel is the greater risk per employee.

4.3 Job Satisfaction



Figure 4: Job Satisfaction

What it shows: Retained and departed employees for each self-reported satisfaction rating (1 to 4).

Finding: Attrition falls as satisfaction rises. Employees who rated their satisfaction as 1 left at 22.8% (66 of 289), compared with 16.5% for rating 3 (73 of 442) and 11.3% for rating 4 (52 of 459).

Insight: Dissatisfied employees are about twice as likely to leave as the most satisfied ones, so satisfaction surveys can work as an early warning signal.

4.4 Salary Analysis

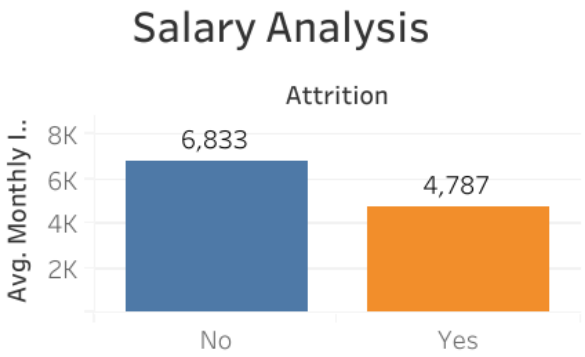


Figure 5: Salary Analysis

What it shows: The average monthly income of employees who stayed versus those who left.

Finding: Employees who stayed earn $6,833 on average, while those who left earned $4,787, about 30% less.

Insight: Lower compensation directly correlates with higher attrition. Underpaid employees are the most exposed to outside offers.

4.5 Commute Impact

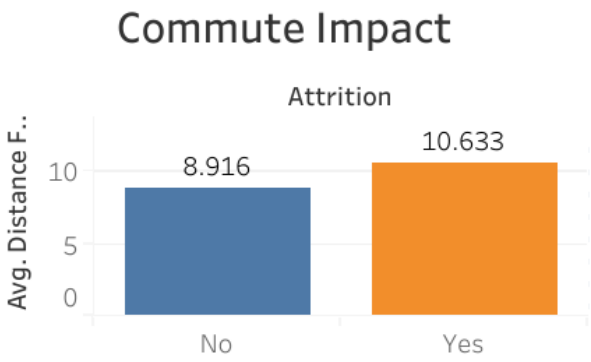


Figure 6: Commute Impact

What it shows: The average distance from home to the office for employees who stayed and for those who left.

Finding: Employees who left lived 10.63 miles away on average, compared with 8.92 miles for those who stayed, a difference of about 1.7 miles.

Insight: A longer daily commute contributes to fatigue and eventual resignation, which makes flexible and hybrid working options worth considering.

4.6 Department Attrition

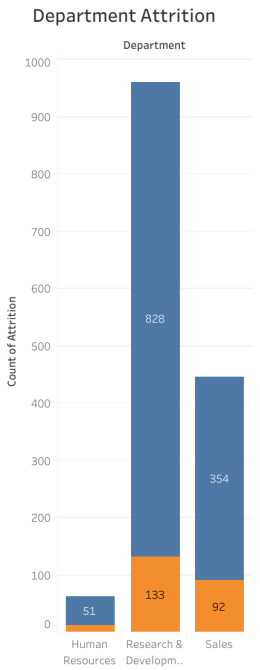


Figure 7: Department Attrition

What it shows: The count of retained and departed employees in Human Resources, Research & Development and Sales.

Finding: R&D has the highest number of departures (133), followed by Sales (92). Because R&D is the largest department, the attrition rate tells a different story: Sales is highest at 20.6%, then Human Resources at 19.0% and R&D at 13.8%.

Insight: R&D loses the most people in absolute terms, while Sales loses the largest share of its team. Both need attention.

5. Segment-Level Attrition Rates

The table below converts the dashboard counts into attrition rates, which allows segments of different sizes to be compared fairly.

Segment	Employees who left	Total employees	Attrition rate
OverTime = Yes	127	416	30.5%
OverTime = No	110	1,054	10.4%
Job Satisfaction = 1 (lowest)	66	289	22.8%
Job Satisfaction = 2	46	280	16.4%
Job Satisfaction = 3	73	442	16.5%
Job Satisfaction = 4 (highest)	52	459	11.3%
Dept: Sales	92	446	20.6%
Dept: Human Resources	12	63	19.0%
Dept: Research & Development	133	961	13.8%
Travel: Frequently	69	277	24.9%
Travel: Rarely	156	1,043	15.0%
Travel: Non-Travel	12	150	8.0%

Overall attrition: 237 of 1,470 employees (16.12%). Counts that are not printed as labels on the dashboard (Human Resources, Travel_Frequently, Non-Travel and Job Satisfaction 2) are taken from the segment totals and reconcile to the 237 total departures.

6. Summary of EDA Findings

- **Overtime** is the strongest driver, with about three times the attrition rate of non-overtime employees.
- **Compensation** matters: leavers earned about 30% less than stayers on average.
- **Job satisfaction** is inversely related to attrition, with the lowest ratings showing about double the attrition of the highest.
- **Commute and travel** add to the risk: leavers lived farther away, and frequent travelers have the highest travel-group attrition rate.
- **Departments:** R&D has the most departures and Sales has the highest rate, so both are priority areas.

> **Next Step**
>
> These findings feed directly into the Business Insights & Strategic Recommendations brief, which turns each driver into a practical action for HR.